

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 4A nr. 43.13

$$\begin{aligned} \int \frac{x^4 + 2x^3 - 12x^2 + 9}{x^3 - 3x + 2} dx &= \int \left(4x + 2 + \frac{-2x + 5}{x^3 - 3x + 2} \right) dx \\ &= 2x^2 + 2x + \int \frac{-2x + 5}{(x+2)(x-1)^2} dx \\ &\rightarrow \frac{-2x + 5}{(x+2)(x-1)^2} = \frac{A}{x+2} + \frac{B}{(x-1)^2} + \frac{C}{x-1} \\ &\Rightarrow A = \frac{4+5}{(-2-1)^2} = \frac{9}{9} = 1 \\ &\Rightarrow B = \frac{-2+5}{3} = 1 \\ &\rightarrow \frac{-2x+5}{(x+2)(x-1)^2} - \frac{1}{(x-1)^2} = \frac{-3}{(x+2)(x-1)} \\ &\Rightarrow C = \frac{-3}{3} = -1 \\ &= 2x^2 + 2x + \int \frac{1}{x+2} dx + \int \frac{1}{(x-1)^2} dx - \int \frac{1}{x-1} dx \\ &= 2x^2 + 2x + \ln|x+2| - \frac{1}{x-1} - \ln|x-1| + c \end{aligned}$$

References